

Sample Problems
Lesson: Probability Distributions
Second Semester, SY 2025 - 2026

General Instructions: Do as instructed.

1. A cybersecurity system checks whether a login attempt is malicious or legitimate. Historical data shows that a login attempt has a probability of 0.08 of being malicious. Let

$$X = \begin{cases} 1, & \text{if the login attempt is malicious} \\ 0, & \text{if the login attempt is legitimate} \end{cases}$$

- (a) What probability distribution is represented by this problem?
 - (b) What is the probability that the login attempt is legitimate?
 - (c) What is the mean and variance of the random variable X ?
2. A software company found that 15% of submitted mobile apps contain at least one critical bug during testing. A quality assurance engineer randomly checks 12 apps. Let X be the number of apps with critical bugs.
- (a) What probability distribution is represented by this problem?
 - (b) What is the probability that exactly 3 apps contain critical bugs?
 - (c) What is the probability that at least 2 apps contain critical bugs?
 - (d) What is the probability that at least 6 apps contain critical bugs?
 - (e) What is the probability that between 4 and 9 apps (inclusive) contain critical bugs?
 - (f) What is the mean and variance of the random variable X ?
3. A cloud server is programmed to automatically start a system backup at a random time between 1:00 AM and 3:00 AM. Assume that every time within this 2-hour interval is equally likely to be selected. Let X represent the number of hours after 1:00 AM when the backup begins.
- (a) What probability distribution is represented by this problem?
 - (b) What is the probability density function of X ?
 - (c) What is the probability that the backup starts after 1 hour?
 - (d) What is the probability that the backup starts between 0.5 and 1.5 hours after 1:00 AM?
 - (e) What are the mean and variance of the distribution?
4. A mobile application company monitors the amount of memory used by its app during execution. The memory consumption is normally distributed with mean 500 MB and standard deviation 50 MB. Let X represent the memory usage (in MB) of the application during runtime.
- (a) What probability distribution is represented by this problem?
 - (b) What is the probability that the application uses at least 575 MB of memory?
 - (c) What is the probability that the application uses between 390 MB and 520 MB of memory?
 - (d) What memory usage corresponds to the lowest 15% of application runs?
 - (e) What are the mean and variance of the distribution?